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Brief Paper

Properties of Least Squares Estimates in Set Membership Identification*†

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Key Words—Worst-case identification; set membership estimation; least squares estimates; unknown but bounded errors.

Abstract—The properties of the least square algorithm in the identification of linear, time-invariant, discrete-time systems, in the presence of unknown but bounded errors are investigated. It is known that the least squares algorithm enjoys strong optimality properties in the case of power bounded (l_2) errors, while it may be far from optimal in the case of pointwise (l_∞) bounded errors. Exact expressions of the local and global worst case errors of the least squares estimates are derived. Using these results, general conditions assuring convergence to zero of the global worst-case error are given. It is also shown that the least squares algorithm may be optimal in l_1 and l_∞ identification of FIR systems with pointwise bounded errors.

1. *Introduction*

The least squares approach, tracing back to Gauss, is certainly the most widely used in system identification and in many other estimation problems. Properties of the least squares algorithm have been extensively investigated in the stochastic setting, showing in particular its strong optimality properties under suitable assumptions on the probability density function of measurement errors.

Much less known are its properties under non-probabilistic assumptions on measurement uncertainties, a case which has been the focus of a growing interest in recent years (see e.g. Schweppe, 1973; Milanese *et al.*, 1989; Milanese and Vicino; 1991; Smith and Dahleh, 1994; and the references cited therein). The main motivation for a set membership approach to system identification is the fact that in many practical cases a bounded error assumption is more realistic and less demanding than statistical assumptions, e.g. in the presence of relevant unmodeled dynamics. Moreover, this approach gives as a result uncertainty models which may be well suited for the use of modern robust control methodologies.

In this paper I investigate the properties of the least square algorithm in such a setting. Kacewicz *et al.* (1986) have shown that the least squares algorithm enjoys strong optimality properties in the case of energy or power bounded (l_2) errors, while it may be far from being optimal in the case of pointwise (l_∞) bounded errors. For this case, some convergence results of least squares estimates of ARX parameters have been derived by Akcay and Khargonekar (1993). The same case has been considered by Gutman (1991), who presented an algorithm for the recursive computation of a bound on the l_∞ error of the least squares estimates.

In this paper I derive exact expressions of the worst case errors of the least squares estimates. These expressions allow one to derive a convergence result which is stronger and more general then the one derived by Akcay and Khargonekar (1993). It is also shown that the least squares algorithm may be optimal in l_1 and l_∞ identification problems with pointwise bounded errors which have been investigated in recent years (see e.g. Mäkilä, 1991; Tse *et al.*, 1991; Chen *et al.*, 1992; Kacewicz and Milanese, 1992; Lin *et al.*, 1992; Dhaleh *et al.*, 1993; Milanese and Kacewicz, 1993; or, for a more complete set of references Milanese and Vicino, 1993).

Note that in this paper we focus on parameter estimation of ARX models. However, this problem is formulated as a general set membership estimation problem (Milanese and Vicino, 1991), and most of the results are obtained as properties of the least squares algorithm in this general context. Then, these results can be applied, directly or with minor modifications, to many other problems, which can be formulated in this general context, such as state estimation, time series prediction, H_∞ identification, worst-case identification with mixed parametric–nonparametric models (Milanese and Tempo, 1985; Helmicki *et al.*, 1991; Gu and Khargonekar, 1992, Lin *et al.*, 1992; Elia and Milanese, 1993; Milanese and Vicino, 1993).

2. *Problem setting*

Let us consider the ARX model

$$y_k = \sum_{i=1}^{l} v_i y_{k-i} + \sum_{i=0}^{m} \theta_i u_{k-i} + e_k, \tag{1}$$

where y_k is the output sequence, u_k is a known input sequence and e_k is a sequence of unknown but bounded errors.

Suppose that $N + l$ values $[y_1, \ldots, y_{N+l}]$ are known and the aim is to estimate the unknown parameters $[v_i, \theta_i]$.

Let Λ be the space of dimension $n = l + m + 1$ with elements:

$$\lambda = [v_1, \ldots, v_l, \theta_0, \ldots, \theta_m]'. \tag{2}$$

Let Y be the N-dimensional space with elements $y^N = [y_{l+1}, \ldots, y_{N+l}]'$. Then, the relation between the measured outputs and the unknown parameters can be written as

$$y^N = F_N \lambda + e^N, \tag{3}$$

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where

$$F_N = \begin{pmatrix} y_l & \cdots & y_1 & u_{l+1} & \cdots & u_{l+1-m} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ y_{N+l-1} & \cdots & y_N & u_{N+l} & \cdots & u_{N+l-m} \end{pmatrix}. \quad (4)$$

The error term $e^N = [e_{l+1}, \ldots, e_{N+l}]^t$ is unknown, but we suppose that it is bounded in some given norm

$$\|e^N\|_Y \leq \epsilon. \quad (5)$$

An *algorithm* ϕ is an operator (in general nonlinear) $\phi: Y \to \Lambda$, providing an approximation $\hat{\lambda} = \phi(y^N)$ of λ. Such an algorithm is also called *estimator*. Let us define the following sets:
Measurement Uncertainty Set: $\mathrm{MUS}_y \doteq \{\tilde{y} \in Y : \|\tilde{y} - y^N\|_N \leq \epsilon\}$.
Feasible Parameter Set: $\mathrm{FPS}_y \doteq \{\lambda : \|y^N - F_N\lambda\|_Y \leq \epsilon\}$.
Estimates Uncertainty Set: $\mathrm{EUS}_y(\phi) \doteq \phi\{\mathrm{MUS}_y\}$.

We suppose that FPS_y is nonempty, i.e. that some ARX models (1) exist, whose outputs differ from measured data no more than ϵ. Otherwise, a priori information on the assumed model structure and on the value of ϵ are not consistent with the measured data, and something has to be changed.

Performance of identification procedures are measured according to worst-case errors. Three different identification errors are considered:

Y-local error: $E_Y^N(\phi, u, \epsilon) \doteq \sup_{\lambda \in \mathrm{FPS}_y} \|\lambda - \phi(y^N)\|_\Lambda$

Λ-local error: $E_\Lambda^N(\phi, u, \epsilon) \doteq \sup_{y^N \in \mathrm{MUS}_{F\lambda}} \|\lambda - \phi(y^N)\|_\Lambda$

Global error:

$$E^N(\phi, u, \epsilon) \doteq \sup_{y^N \in Y} E_Y^N(\phi, u, \epsilon) = \sup_{\lambda \in \Lambda} E_\Lambda^N(\phi, u, \epsilon).$$

The Y-local error is also called a posteriori *worst-case error*, being the error that can be actually guaranteed, after the measurements y^N are made available. For a discussion on the interpretation of these errors see, e.g. Milanese and Vicino (1991).

Algorithms minimizing these types of errors are called *Y-locally optimal*, *Λ-locally optimal* and *globally optimal*, respectively. Note that if an algorithm is Y- or Λ-locally optimal, then it is also globally optimal (Milanese and Tempo, 1985), while the converse is not true.

Properties of algorithms are also typically discussed in terms of convergence to zero of the identification error. Several convergence concepts have been introduced in the literature, using sometimes the same names for different definitions.

In this paper an identification procedure is called *convergent* if:

$$\lim_{\epsilon \to 0} \lim_{N \to \infty} E_\Lambda^N(\phi, u, \epsilon) = 0, \quad \forall \lambda \in \Lambda. \quad (6)$$

This is the convergence concept used by Akcay and Khargonekar (1993). Sometimes it is indicated as robust convergence, e.g. by Mäkilä (1991).

An algorithm is called globally *convergent* if:

$$\lim_{\epsilon \to 0} \lim_{N \to \infty} E^N(\phi, u, \epsilon) = 0. \quad (7)$$

This concept is indicated sometimes as uniform convergence, (e.g. Tse *et al.*, 1991). Global convergence implies convergence, while the converse is not true.

The *least squares algorithm* ϕ^{LS} is defined as

$$\phi^{\mathrm{LS}}(y) = \lambda_Y^{\mathrm{LS}}, \quad \lambda_Y^{\mathrm{LS}} = \arg\min_\lambda \|y^N - F_N\lambda\|_2. \quad (8)$$

Note that the more general case of weighted least squares could be considered as well. The present formulation is taken only for the sake of notational simplicity.

Throughout the paper we suppose that rank $[F_N] = n$. This is called the *sufficient information hypothesis*, since, if it does not hold, FPS_y is unbounded and consequently the identification errors $E_Y^N(\phi, u, \epsilon)$ and $E^N(\phi, u, \epsilon)$ are not finite.

Under the sufficient information hypothesis, $\phi^{\mathrm{LS}}(y)$ is unique and linear, and is given by

$$\phi^{\mathrm{LS}}(y) = (F_N^t F_N)^{-1} F_N^t y^N = A y^N. \quad (9)$$

Note that the least squares estimation matrix A depends on input u. When necessary, such a dependence will be evidenced by the notation A_u.

All input, output and noise sequences are considered in this paper as deterministic signals. We will make use of concepts such as *quasi-stationarity and persistent excitation*, which are very similar to analogous (and more familiar) concepts for stochastic signals. For example, a sequence u is quasi-stationary if it is bounded and the limit $R_u(\tau) = \lim_{N \to \infty} (1/N) \sum_{k=1}^{N} u_k u_{k-\tau}$, exists. We refer to Ljung (1987) for formal definitions.

3. *Identification error evaluation*

The sets PFS_y and $\mathrm{EUS}_y(\phi^{\mathrm{LS}})$ can be conveniently described in term of the least squares estimation matrix A, as shown in the following result: here $\mathcal{R}(F_N)$ indicates the range space of F_N.

Theorem 1.
(i) $\mathrm{FPS}_y = A\{\mathrm{MUS}_y \cap \mathcal{R}(F_N)\}$;
(ii) $\mathrm{EUS}_y(\phi^{\mathrm{LS}}) = \lambda_y^{\mathrm{LS}} \oplus A\{\mathrm{MUS}_0\}$.

Proof. Let us first prove that $\tilde{\lambda} \in \mathrm{FPS}_y \Rightarrow \tilde{\lambda} \in A\{\mathrm{MUS}_y \cap \mathcal{R}(F_N)\}$. Clearly, if $\tilde{\lambda} \in \mathrm{FPS}_y$, then $\tilde{y} = F_N\tilde{\lambda} \in \mathrm{MUS}_y \cap \mathcal{R}(F_N)$. Now, $A\tilde{y} = (F_N^t F_N)^{-1} F_N^t F_N \tilde{\lambda} = \tilde{\lambda}$, which implies that $\tilde{\lambda} \in A\{\mathrm{MUS}_y \cap \mathcal{R}(F_N)\}$. Now we prove that $\tilde{\lambda} \in A\{\mathrm{MUS}_y \cap \mathcal{R}(F_N)\} \Rightarrow \tilde{\lambda} \in \mathrm{FPS}_y$. If $\tilde{\lambda} \in A\{\mathrm{MUS}_y \cap \mathcal{R}(F_N)\}$, then $\tilde{\lambda} = A\tilde{y}$ for some $\tilde{y} \in \mathrm{MUS}_y \cap \mathcal{R}(F_N)$. Since $\tilde{y} \in \mathcal{R}(F_N)$, then $\tilde{y} = F_N\tilde{\lambda}$. However, $\tilde{\lambda} = A\tilde{y} = (F_N^t F_N)^{-1} F_N^t F_N \tilde{\lambda} = \tilde{\lambda}$. Then it results that $\tilde{y} = F_N\tilde{\lambda}$. Since also $\tilde{y} \in \mathrm{MUS}_y$, then $\tilde{\lambda} \in \mathrm{FPS}_y$. Thus we have proved (i). Part (ii) of the theorem follows from the fact that $\mathrm{MUS}_y = y \oplus \mathrm{MUS}_0$ and the linearity of ϕ^{LS}. ■

Remark 1. From (i) of Theorem 1 it follows that:

$$\mathrm{FPS}_y \subseteq \mathrm{EUS}_y(\phi^{\mathrm{LS}}), \quad \forall y. \quad (10)$$

This is true for any *correct algorithm*, i.e. for any ϕ such that $\phi(F_N\lambda) = \lambda$ (Milanese and Belforte, 1982). ■

Using Theorem 1, the following expressions of the identification errors can be derived. We recall that the induced norm of a matrix $A: Y \to \Lambda$ is:

$$\|A\|_{Y,\Lambda} \doteq \sup_{\|y\|_Y \leq 1} \|Ay\|_\Lambda. \quad (11)$$

Theorem 2.

$$E_Y^N(\phi^{\mathrm{LS}}, u, \epsilon) \leq E_\Lambda^N(\phi^{\mathrm{LS}}, u, \epsilon) = E^N(\phi^{\mathrm{LS}}, u, \epsilon) = \epsilon \|A\|_{Y,\Lambda}, \quad \forall y, \lambda.$$

Proof. From the definition of $E_\Lambda^N(\phi^{\mathrm{LS}}, u, \epsilon)$ it follows:

$$E_\Lambda^N(\phi^{\mathrm{LS}}, u, \epsilon) = \sup_{y^N \in \mathrm{MUS}_{F\lambda}} \|\lambda - \phi^{\mathrm{LS}}(y^N)\|_\Lambda = \sup_{\tilde{\lambda} \in \mathrm{EUS}} \|\lambda - \tilde{\lambda}\|_\Lambda. \quad (12)$$

From Theorem 1 we have

$$\tilde{\lambda} \in \mathrm{EUS}_{F\lambda} \Rightarrow \tilde{\lambda} = \lambda_{F_N\lambda}^{\mathrm{LS}} + \bar{\lambda}, \quad \bar{\lambda} \in \mathrm{EUS}_0. \quad (13)$$

Since

$$\lambda_{F\lambda}^{\mathrm{LS}} = AF\lambda = (F_N^t F_N)^{-1} F_N^t F_N \lambda = \lambda, \quad (14)$$

we have

$$\tilde{\lambda} \in \mathrm{EUS}_{F\lambda} \Rightarrow \tilde{\lambda} = \lambda + \bar{\lambda}, \quad \bar{\lambda} \in \mathrm{EUS}_0. \quad (15)$$

Then

$$\sup_{\tilde{\lambda} \in \mathrm{EUS}_{F\lambda}} \|\lambda - \tilde{\lambda}\|_\Lambda = \sup_{\bar{\lambda} \in \mathrm{EUS}_0} \|\bar{\lambda}\|_\Lambda = \sup_{\bar{y} \in \mathrm{MUS}_0} \|A\bar{y}\|_\Lambda = \sup_{\|\bar{y}\| \leq \epsilon} \|A\bar{y}\|_\Lambda = \epsilon \|A\|_{Y,\Lambda}. \quad (16)$$

Thus, $E_\lambda^N(\phi^{LS}, u, \epsilon) = \epsilon \|A\|_{Y,\Lambda}$ independent of λ. The result (Milanese and Tempo, 1985):

$$\sup_{\lambda \in \Lambda} E_\lambda^N(\phi, u, \epsilon) = E^N(\phi, u, \epsilon), \quad \forall \phi \tag{17}$$

and $E_y^N(\phi^{LS}, u, \epsilon) \le E^N(\phi^{LS}, u, \epsilon)$ complete the proof. ■

Remark 2. Theorem 2 is true for any linear and correct ϕ. In particular it is true for any algorithm of the form $\phi^W(y^N) = (F_N' W F_N)^{-1} F_N' W y^N$, with W symmetric and positive definite. ■

In the literature, the cases of pointwise bounded errors and power bounded errors are of most concern. This consists in assuming $\|\cdot\|_Y = \|\cdot\|_\infty$ in the first case and $\|\cdot\|_Y = (1/\sqrt{N})\|\cdot\|_2$ in the second case. Concerning the Λ space, $\|\cdot\|_p$ norms are typically considered, with $p = 1, 2, \infty$.

Exact expressions of $\|A\|_{Y,\Lambda}$ are known for some of these norms in Y and Λ and are reported in Table 1 (Dunford and Schwarz, 1958; Ortega, 1987).

To the best of the author's knowledge, no expressions are known for $\|A\|_{2,1}$, $\|A\|_{\infty,1}$ and $\|A\|_{\infty,2}$. These quantities can be computed by solving (11) directly, using a convex optimization program. Alternatively, upper and lower bounds can be obtained by means of standard relations among norms.

For example, Akcay and Khargonekar (1993) investigated the Λ-local error in the case $\|\cdot\|_Y = \|\cdot\|_\infty$, $\|\cdot\|_\Lambda = \|\cdot\|_2$. Expression of $\|A\|_{\infty,2}$ is not known, but since $\|\lambda\|_\infty \le \|\lambda\|_2 \le \sqrt{n}\,\|\lambda\|_\infty$, it follows:

$$\|A\|_{\infty,\infty} \le \|A\|_{\infty,2} \le \sqrt{n}\,\|A\|_{\infty,\infty} \tag{18}$$

$$\|A\|_{2,2} \le \|A\|_{\infty,2} \le \sqrt{n}\,\|A\|_{2,2}. \tag{19}$$

From these bounds, Table 1 and Theorem 2 we derive the following result:

Proposition 1. If $\|\cdot\|_Y = \|\cdot\|_\infty$, and $\|\cdot\|_\Lambda = \|\cdot\|_2$, then:

$$\begin{aligned}\epsilon \max\left\{\bar\sigma(A), \max_i \sum_{j=1}^N |a_{ij}|\right\} &\le E_\lambda^N(\phi^{LS}, u, \epsilon)\\ &\le \epsilon\sqrt{n}\min\left\{\bar\sigma(A), \max_i \sum_{j=1}^N |a_{ij}|\right\}, \quad \forall u.\end{aligned}$$

Remark 3. Proposition 1 represents a significant improvement over the main result of Akcay and Khargonekar (1993), who gave a less tight upper bound of $E_\lambda^N(\phi^{LS}, u, \epsilon)$, and under the assumptions that the system to be identified is exponentially stable with decay rate and gain bounded by known constants, the input is a pseudo random binary sequence and N is sufficiently large. ■

As the a posteriori error $E_y^N(\phi^{LS}, u, \epsilon)$ is concerned, Theorem 2 gives only its maximal achievable value for all admissible measurement errors. However, its actual value is clearly of interest, and the evaluation of $E_y^N(\phi^{LS}, u, \epsilon)$ is now investigated.

From Theorem 1 it follows that, if $\|\cdot\|_Y = \|\cdot\|_\infty$, the sets FPS_y and $EUS_y(\phi_{LS})$ are convex polytopes. In fact, MUS_y is a convex polytope (actually, a hypercube). Intersection of a polyhedron with an hyperplane results in a polytope, and mapping a polytope through a linear operator gives a polytope. Then, due to convexity of FPS_y, the Y-local identification error can be computed from the vertices $\lambda^1, \lambda^2, \ldots, \lambda^r$ of FPS_y, as given by the following result:

Table 1. Expressions of $\|A\|_{Y,\Lambda} \cdot \bar\sigma(A)$: maximal singular value of A

$Y \downarrow \Lambda \rightarrow$	l_1	l_2	l_∞
l_1	$\max_j \sum_i^n \lvert a_{ij}\rvert$	$\max_j \sum_i^n \{a_{ij}^2\}^{1/2}$	$\max_{i,j} \lvert a_{ij}\rvert$
l_2		$\bar\sigma(A)$	$\max_i \sum_j^N \{a_{ij}^2\}^{1/2}$
l_∞			$\max_i \sum_j^N \lvert a_{ij}\rvert$

Proposition 2. If $\|\cdot\|_Y = \|\cdot\|_\infty$, then:

$$E_y^N(\phi^{LS}, u, \epsilon) = \max_{i=1,\ldots,r} \|\lambda^i - \lambda_{LS}\|_\Lambda,$$

where λ^i is the i-th vertex of FPS_y. ■

Remark 4. The computation of the a posteriori identification error through Proposition 2 requires the computation of the vertices of FPS_y. Algorithms for computing the vertices of a polytope can be found, (e.g. in Matheiss and Rubin, 1980; Mo and Norton, 1988; Piet-Lahanier and Walter, 1989). ■

If $\|\cdot\|_Y = \|\cdot\|_2$, the sets FPS_y and $EUS_y(^{LS})$ are ellipsoids. Using standard least squares arguments, the feasible parameter set can be expressed as (Schweppe, 1973; Wahlberg and Ljung, 1992):

$$\begin{aligned}FPS_y = \{\lambda \in \Lambda : &(\lambda - \lambda_y^{LS})'(F_N' F_N)(\lambda - \lambda_y^{LS})\\ &\le \epsilon^2 - \epsilon_r^2\},\end{aligned} \tag{20}$$

where

$$\epsilon_r^2 = (y^N - F_N\lambda_y^{LS})'(y^N - F_N\lambda_y^{LS}).$$

Note that $\epsilon_r \le \epsilon$, since ϵ_r is the minimum distance of y^N from $\mathcal{R}(F_N)$ and FPS_y is supposed nonempty, i.e. $\|y^N - F_N\lambda\|_2 \le \epsilon$ for some λ.

In the next theorem we derive expressions for $EUS_y(\phi^{LS})$ and for the a posteriori identification error.

Theorem 3. Let $\|\cdot\|_Y = \|\cdot\|_2$. Then:
(i) $EUS_y(\phi^{LS}) = \{\lambda \in \Lambda : (\lambda - \lambda_y^{LS})'(F_N' F_N)(\lambda - \lambda_y^{LS}) \le \epsilon^2\}$;
(ii) $E_y^N(\phi^{LS}, u, \epsilon) = \sqrt{\epsilon^2 - \epsilon_r^2}\,\|A\|_{2,\Lambda}$.

Proof. In order to prove (i), recall that $MUS_y = \{\tilde y \in Y : (\tilde y - y^N)'(\tilde y - y^N) \le \epsilon^2\}$ and $EUS_y(\phi^{LS}) = A\{MUS_y\}$.

Using a standard result on linear transformation of ellipsoids (Schweppe, 1973), it results:

$$\begin{aligned}EUS_y(\phi^{LS}) = \{\lambda \in \Lambda : &(\lambda - Ay^N)'(AA')^{-1}\\ &\times (\lambda - Ay^N) \le \epsilon^2\}.\end{aligned} \tag{21}$$

Then (i) follows recalling that $Ay^N = \lambda_y^{LS}$ and observing that:

$$AA' = (F_N' F_N)^{-1} F_N' F_N ((F_N' F_N)^{-1})' = (F_N' F_N)^{-1}. \tag{22}$$

To prove (ii) we note that:

$$E_y^N(\phi^{LS}, u, \epsilon) = \{\max \|\lambda\|_\Lambda : \lambda'(F_N' F_N)\lambda \le \epsilon^2 - \epsilon_r^2\} \tag{23}$$

$$E_\lambda^N(\phi^{LS}, u, \epsilon) = \{\max \|\lambda\|_\Lambda : \lambda'(F_N' F_N)\lambda \le \epsilon^2\} \tag{24}$$

so that

$$E_y^N(\phi^{LS}, u, \epsilon) = E_\lambda^N(\phi^{LS}, u, \epsilon)\sqrt{(\epsilon^2 - \epsilon_r^2)}/\epsilon. \tag{25}$$

Then (ii) follows, recalling from Theorem 2 that:

$$E_\lambda^N(\phi^{LS}, u, \epsilon) = \epsilon\,\|A\|_{2,\Lambda}. \tag{26}$$

■

4. *Optimality properties*

In this section we investigate the optimality properties of the least squares algorithm. The strongest concept related to optimal algorithms is the one of the central algorithm (Traub *et al.*, 1988). A *central algorithm* ϕ^c is defined as

$$\phi^c(y) \doteq c(FPS_y), \tag{27}$$

where $c(FPS_y)$ is the (Chebicheff) center of FPS_y, given by

$$\sup_{\lambda \in FPS_y} \|c(FPS_y) - \lambda\|_\Lambda = \inf_{\tilde\lambda} \sup_{\lambda \in FPS_y} \|\tilde\lambda - \lambda\|_\Lambda. \tag{28}$$

A central algorithm is Y-locally and globally optimal (Traub *et al.*, 1988). In particular it gives the minimal global identification error, denoted as $R^N(u, \epsilon)$ and called the *radius of information*:

$$R^N(u, \epsilon) \doteq \inf_\phi E^N(\phi, u, \epsilon) = E^N(\phi^c, u, \epsilon). \tag{29}$$

Kacewicz *et al.* (1986) have shown that if Y is an Hilbert space, then ϕ^{LS} is Y-locally and Λ-locally optimal (and consequently globally optimal), whatever norm is assumed in the parameter space Λ. On the contrary, if $\|\cdot\|_Y = \|\cdot\|_\infty$, in general ϕ^{LS} is not optimal in any of the defined concepts (see e.g. Examples 1 and 2 below). However, we now show that ϕ^{LS} is Λ-locally optimal for some identification problem with pointwise bounded errors (i.e. $\|e^N\|_\infty \le \epsilon$), which has been recently investigated in the literature, namely worst-case l_1 and l_∞ identification (see e.g. Mäkilä, 1991; Tse *et al.*, 1991; Chen *et al.*, 1992; Kacewicz and Milanese, 1992; Lin *et al.*, 1992; Dahleh *et al.*, 1993; Milanese and Kacewicz, 1993). Note that, for the sake of brevity, only FIR identification is considered here, though these results can be easily extended to nonparametric identification of exponentially stable infinite impulse response linear systems.

The quantity $\|e\|_\infty/\|u\|_\infty$ can be interpreted as a noise-to-signal ratio. Since all are linear, without loss of generality, in the next theorems we consider inputs with unitary $\|\cdot\|_\infty$ norm. In these theorems the unit impulse and step sequences are denoted as δ and s, respectively.

Theorem 4. (*FIR l_1 identification.*) Let $l=0$, $m>1$, $\|\cdot\|_Y = \|\cdot\|_\infty$ and $\|\cdot\|_\Lambda = \|\cdot\|_1$. Then ϕ^{LS} is Λ-locally optimal when δ or s are used as inputs. Moreover:
(i) $R^N(\delta, \epsilon) = E^N(\phi^{LS}, \delta, \epsilon) = (m+1)\epsilon$;
(ii) $R^N(s, \epsilon) = E^N(\phi^{LS}, s, \epsilon) = (2m+1)\epsilon$.

Proof. The least squares estimation matrices A_δ and A_s have the following expressions:

$$A_\delta = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & \cdot & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & \cdot & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & 1 & 0 & \cdot & 0 \end{pmatrix} \tag{30}$$

$$A_s = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & \cdot & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & \cdot & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & -1 & (N-m)^{-1} & \cdot & (N-m)^{-1} \end{pmatrix}. \tag{31}$$

From these expressions, it is easily seen that:

$$\|A_\delta\|_{\infty,1} = m+1 \tag{32}$$

$$\|A_s\|_{\infty,1} = 2m+1. \tag{33}$$

Then from Theorem 2 it follows:

$$E^N(\phi^{LS}, \delta, \epsilon) = (m+1)\epsilon \tag{34}$$

$$E^N(\phi^{LS}, s, \epsilon) = (2m+1)\epsilon. \tag{35}$$

On the other hand, from a basic result in Information Based Complexity (Traub *et al.*, 1988, Lemma 5.2.1), we have:

$$R^N(u, \epsilon) \ge \sup_{\theta \in FPS_0} \|\theta\|_1. \tag{36}$$

It is easy to see that:

$$\begin{aligned} R^N(\delta, \epsilon) &\ge \sup\{\|\theta\|_1 : |\theta_j| \le \epsilon, j = 0, 1, \ldots, m\} \\ &\ge (m+1)\epsilon \end{aligned} \tag{37}$$

$$\begin{aligned} R^N(s, \epsilon) &\ge \sup\{\|\theta\|_1 : |\theta_0 + \cdots + \theta_j| \\ &\qquad \le \epsilon, j = 0, 1, \ldots, m\} \\ &\ge (2m+1)\epsilon. \end{aligned} \tag{38}$$

Comparing with (34) and (35) and recalling that $R^N(u, \epsilon) \le E^N(\phi, u, \epsilon)$, $\forall \phi$, $\forall u$, one obtains:

$$R^N(\delta, \epsilon) = E^N(\phi^{LS}, \delta, \epsilon) \tag{39}$$

$$R^N(s, \epsilon) = E^N(\phi^{LS}, s, \epsilon) \tag{40}$$

which prove the theorem. ■

From theorem 4 it results that, using impulse or step as inputs, the global l_1 identification error cannot be less than $(m+1)\epsilon$ and $(2m+1)\epsilon$, respectively, whatever the identification algorithm and the value of N are. On the other hand, it has been shown that the global identification error can be made as small as 2ϵ, provided that Galois sequences are used as input, $N = O(2^{m+1})$, and an interpolatory algorithm is used (Mäkilä, 1991; Kacewicz and Milanese, 1992). It is not known if such a value can be obtained by means of the least squares algorithm. Note that in the noise free case, impulsive input is optimal, i.e. gives the minimal radius of information (Lin *et al.*, 1992), but this is no more true in the noise corrupted case.

On the contrary, the impulse is an optimal input in l_∞ noisy identification. In this case, Milanese and Vicino (1993) have shown that:

$$\inf_u R^N(u, \epsilon) = \epsilon. \tag{41}$$

This minimal error can be obtained from the least squares algorithm applied to impulse response data, as shown by the next theorem.

Theorem 5. (*l_∞ identification.*) Let $l=0$, $m>1$, $\|\cdot\|_Y = \|\cdot\|_\infty$ and $\|\cdot\|_\Lambda = \|\cdot\|_\infty$. Then, ϕ^{LS} is optimal when δ or s are used as inputs. Moreover:
(i) $R^N(\delta, \epsilon) = E^N(\phi^{LS}, \delta, \epsilon) = \epsilon$;
(ii) $R^N(s, \epsilon) = E^N(\phi^{LS}, s, \epsilon) = 2\epsilon$.

Proof. From (30) and (31) it follows that:

$$\|A_\delta\|_{\infty,\infty} = 1 \tag{42}$$

$$\|A_s\|_{\infty,\infty} = 2. \tag{43}$$

On the other hand it results:

$$R^N(\sigma, \epsilon) \ge \sup\{\|\theta\|_\infty : |\theta_j| \le \epsilon, j = 0, 1, \ldots, m\} \ge \epsilon \tag{44}$$

$$\begin{aligned} R^n(s, \epsilon) &\ge \sup\{\|\theta\|_\infty : |\theta_0 + \cdots + \theta_j| \\ &\qquad \le \epsilon, j = 0, 1, \ldots, m\} \ge 2\epsilon. \end{aligned} \tag{45}$$

Following the argument of Theorem 4, the claim follows. ■

In the case of pointwise bounded errors, least squares estimates and central estimates may have the same global error, as shown in Theorems 5 and 6. However, the central estimate has lower a posteriori worst-case error, since ϕ^c is Y-locally optimal, while ϕ^{LS} is not. Indeed, the ratio $E_y^N(\phi^{LS}, u, \epsilon)/E_y^N(\phi^c, u, \epsilon)$ may be unbounded, as shown by the following example:

Example 1. Suppose that the data $y_1 = 1.2$, $y_2 = 0.3$ are generated by the FIR system:

$$y_k = \lambda u_k + e_k, \tag{46}$$

with $u_1 = 1$, $u_2 = 0.5$, $|e_k| \le 0.2$. It results:

$$\begin{aligned} FPS_y &= \{\lambda : |1.2 - \lambda| \le 0.2, |0.3 - 0.5\lambda| \le 0.2\} \\ &= \{\lambda : \lambda = 1\} \end{aligned} \tag{47}$$

$$\lambda_y^{LS} = 1.08. \tag{48}$$

Then, by considering $\|\cdot\|_\infty$ norm in both Y and Λ spaces, one obtains:

$$E_y^N(\phi^{LS}, u, \epsilon) = 0.08 \tag{49}$$

$$E_y^N(\phi^c, u, \epsilon) = 0.0. \tag{50}$$

Note that the least squares esitmate does not belong to the feasible parameter set. ■

This example shows that, in the pointwise bounded case, it may be worthwhile to use a central algorithm, instead of the least squares one, even when ϕ^{LS} is globally optimal. Constructing central algorithms is in general not an easy task, though solutions can be found for some interesting classes of problems, see e.g. (Milanese and Tempo, 1985; Traub *et al.*, 1988). The concept of interpolatory algorithms can be used to construct almost optimal algorithms (i.e. optimal within a

factor of 2), which can be computed more easily. An *interpolatory algorithm* ϕ^I is defined as

$$\phi^I(y) = \lambda_y^I, \quad \lambda_y^I \in \mathrm{FPS}_y. \tag{51}$$

Interpolatory algorithms are almost y-locally optimal, since it holds (Traub *et al.*, 1988):

$$E_y^N(\phi^I, u, \epsilon) \le 2E_y^N(\phi^c, u, \epsilon) \le 2 \inf_{\phi} E_y^N(\phi, u, \epsilon) \forall \phi. \tag{52}$$

Interpolatory estimates can be easily computed by solving one linear programming problem (Vicino and Milanese, 1991), and may have much lower a posteriori error than least squares estimates. Indeed, the ratio $E_y^N(\phi^{\mathrm{LS}}, u, \epsilon)/E_y^N(\phi^I, u, \epsilon)$ may be unbounded, as shown by the previous example.

5. *Convergence properties*

At first, we present an example where the least squares algorithm is not convergent.

Example 2. Suppose that the data are generated by the FIR system:

$$y_k = \lambda u_k + e_k. \tag{53}$$

Clearly $F'_N = [u_1 u_2 \cdots u_N]$, so that:

$$A = \frac{[u_1 u_2 \cdots u_N]}{\|u\|_2^2}. \tag{54}$$

By considering $\|\cdot\|_Y = \|\cdot\|_\infty$ and $\|\cdot\|_\Lambda = \|\cdot\|_\infty$ it results from Theorem 2 and Table 1:

$$E_\lambda^N(\phi^{\mathrm{LS}}, u, \epsilon) = \epsilon \, \|u\|_1 / \|u\|_2^2. \tag{55}$$

If

$$u = u^* = \{1, 1/\sqrt{2}, \ldots, 1/\sqrt{N}\} \tag{56}$$

it results:

$$E_\lambda^N(\phi^{\mathrm{LS}}, u^*, \epsilon) = \epsilon \frac{\sum_{k=1}^N 1/\sqrt{k}}{\sum_{k=1}^N 1/k}. \tag{57}$$

It can easily be seen that the series $\left(\sum_{k=1}^N 1/\sqrt{k}\right) \Big/ \left(\sum_{k=1}^N 1/k\right)$ is divergent, so that:

$$\lim_{N\to\infty} E_\lambda^N(\phi^{\mathrm{LS}}, u^*, \epsilon) = \infty. \tag{58}$$

Then, using the input (56), the least squares algorithm is not convergent. Note that, on the contrary, the error of the central algorithm is not divergent. In fact, it can be easily seen (Milanese and Tempo, 1985, Theorem 4) that:

$$E_\lambda^N(\phi^c, u^*, \epsilon) \le E^N(\phi^c, u^*, \epsilon) = \epsilon, \quad \forall N. \tag{59}$$

Note also that if $u = \delta = [1, 0, \ldots, 0]$, then:

$$E_\lambda^N(\phi^{\mathrm{LS}}, \delta, \epsilon) = \epsilon, \tag{60}$$

so that with impulse input the least squares algorithm is convergent. ■

In the previous example, the identification error of ϕ^{LS} is divergent as $N \to \infty$, while the error of the central algorithm is bounded. Indeed, the next result shows that, for pointwise bounded errors, divergence of the estimate's error never occurs for central or interpolatory algorithms.

Theorem 6. Let $\|\cdot\|_Y = \|\cdot\|_\infty$. Then, the central algorithm ϕ^c and interpolatory algorithm ϕ^I are globally convergent.

Proof. From a basic result in Information Based Complexity, (Traub *et al.*, 1988, Lemma 5.2.1), we have:

$$E^N(\phi^c, u, \epsilon) \le E^N(\phi^I, u, \epsilon) \le D^N(u, \epsilon), \tag{61}$$

where

$$D^N(u, \epsilon) = 2 \sup_{\lambda \in \mathrm{FPS}_0} \|\lambda\|_\Lambda. \tag{62}$$

Then, to prove the theorem it suffices to show that $\lim_{\epsilon\to 0} \lim_{N\to\infty} D^N(u, \epsilon) = 0$.

Now as new measurements are added, the feasible parameter sets are successively nested, i.e.

$$\mathrm{FPS}_0^{N+1} \subseteq \mathrm{FPS}_0^N. \tag{63}$$

From the sufficient information hypothesis, it follows that there exists N_0 such that $\mathrm{FPS}_0^{N_0}$ is bounded. Then, $D^N(u, \epsilon)$ is a non-increasing function of N and bounded as:

$$D^N(u, \epsilon) \le D^{N_0}(u, \epsilon), \quad \forall N \ge N_0. \tag{64}$$

From definition (62) it is clear that $\lim_{\epsilon\to 0} D^{N_0}(u, \epsilon) = 0$. Then, from (64), we have

$$\lim_{\epsilon\to 0} \lim_{N\to\infty} D^N(u, \epsilon) = 0 \tag{65}$$

which proves the theorem. ■

Example 2 shows that some conditions have to be given in order to assure convergence of the least squares algorithm. The next theorem gives quite general sufficient conditions. Note that from Theorem 2 it follows that if ϕ^{LS} is convergent, it is globally convergent also.

Theorem 7. Let $\|\cdot\|_Y = \|\cdot\|_\infty$ or $\|\cdot\|_Y = (1/\sqrt{N})\|\cdot\|_2$, u and e be quasi-stationary, u be persistently exciting and the unknown system be stable. Then, ϕ^{LS} is globally convergent.

Proof. Since $A = (F'_N F_N)^{-1} F'_N$, it results (Ortega, 1987, p. 94):

$$\|A\|_{Y,\Lambda} \le \|(F'_N F_n)^{-1}\|_{\infty,\Lambda} \, \|F'_N\|_{Y,\infty}. \tag{66}$$

The matrix $F'_N F_N$ has elements of the type $\sum_{k=1}^N x_k z_{k-\tau}$, where x and z may be u or y. Since y is obtained by quasi-stationary signals u and e filtered by a stable system, it is itself quasi-stationary and $\lim_{N\to\infty} (1/N) \sum_{k=1}^N x_k z_{k-\tau}$ exists. Moreover, if u is persistently exciting, the matrix

$$R_N = \frac{1}{N} F'_N F_N \tag{67}$$

is uniformly (with respect to N) positive definite (Ljung, 1987). This, together with the sufficient information hypothesis, implies that the elements $\bar{r}_{ij}^N$ of R_N^{-1} can be bounded by a constant M_R independent of N:

$$|\bar{r}_{ij}^N| \le M_R. \tag{68}$$

From the Norm Equivalence Theorem (Ortega, 1987, p. 90), we have:

$$\|R_N^{-1}\|_{\infty,\Lambda} \le c \, \|R_N^{-1}\|_{1,\infty}, \tag{69}$$

where c is a constant not dependent on N. Then:

$$\|(F'_N F_N)^{-1}\|_{\infty,\Lambda} = (1/N) \, \|R_N^{-1}\|_{\infty,\Lambda} \le (1/N) c \max_{i,j} |\bar{r}_{ij}^N| \le c M_R / N. \tag{70}$$

Now we derive a bound on $\|F'_N\|_{Y,\infty}$. Since u and e are bounded and the data are generated by a stable system, y is bounded. Then, the elements f_{ij}^N of F_N are bounded by a constant M_F not dependent on N:

$$|f_{ij}^N| \le M_F. \tag{71}$$

From Table 1 it results:

$$\|F'_N\|_{1,\infty} = \sup_{y^N} \|F'_N y^N\|_\infty / \|y^N\|_1 = \max_{i,j} |f_{ij}^N| \le M_F. \tag{72}$$

In the case $\|\cdot\|_Y = (1/\sqrt{N})\|\cdot\|_2$, recalling that $\|y^N\|_2 \ge (1/\sqrt{N})\|y^N\|_1$, we have

$$\begin{aligned} \|F'_N\|_{Y,\infty} &= \sup_{y^N} \sqrt{N} \, \|F'_N y^N\|_\infty / \|y^N\|_2 \\ &\le \sup_{y^N} N \, \|F'_N y^N\|_\infty / \|y^N\|_1 \\ &\le N M_F. \end{aligned} \tag{73}$$

In the case $\|\cdot\|_Y = \|\cdot\|_\infty$, recalling that $\|y^N\|_\infty \geq (1/N)\|y^N\|_1$, we have:

$$\begin{aligned}\|F_N'\|_{\Lambda,\infty} &= \sup_{y^N} \|F_N' y^N\|_\infty / \|y^N\|_\infty \\ &\leq \sup_{y^N} N \|F_N' y^N\|_\infty / \|y^N\|_1 \\ &\leq N M_F. \end{aligned} \tag{74}$$

Finally, from Theorem 2 we have:

$$\begin{aligned} E^N(\phi^{LS}, u, \epsilon) &= \epsilon \|A\|_{Y,\Lambda} \leq \epsilon \|(F_N' F_N)^{-1}\|_{\infty,\Lambda} \|F_N'\|_{Y,\infty} \\ &\leq \epsilon c M_K M_F. \end{aligned} \tag{75}$$

Since this bound is independent of N, we have:

$$\lim_{\epsilon \to 0} \lim_{N \to \infty} E^N(\phi^{LS}, u, \epsilon) = 0. \tag{76}$$

■

Remark 5. Akcay and Khargonekar (1993) have given sufficient conditions for the convergence of the least squares algorithm in the case that $\|\cdot\|_Y = \|\cdot\|_\infty$, $\|\cdot\|_\Lambda = \|\cdot\|_2$, and under the assumptions that the input is a pseudo-random binary sequence and the system to be identified is exponentially stable with known bounds on the gain and decay rates. Theorem 7 represents a more general and stronger convergence result. In fact, it holds for a more general class of inputs (recall that pseudo-random binary sequences are persistently exciting) and does not require any knowledge on the degree of stability. Moreover, the technique used by Akcay and Khargonekar (1993) did not allow the proof of global convergence, which, on the contrary, can be guaranteed by Theorem 7.

Remark 6. The conditions in Theorem 7 (and in particular persistent excitation) are not necessary for convergence. In fact, from Theorems 4 and 5 it follows that, for FIR system identification, ϕ^{LS} is convergent when impulse or step inputs are used, though these inputs are not persistently exciting.

5. *Conclusions*

In the paper I have studied optimality and convergence properties of the least squares estimates in the presence of unknown but bounded errors. I have shown that the least squares algorithm may have interesting optimality properties in the bounded error setting. I have also shown that the error of least squares estimates may diverge as the number of measurements grows, and quite general conditions assuring convergence are derived.

Some discussions on the relationships between least squares and central or interpolatory estimates may be in order. It is known that the least squares algorithm is optimal with respect to any of the considered worst case errors in the case of power bounded errors, while it may be far from optimality in the case of pointwise bounds. However, I have shown that, in the latter case, the least squares estimates are globally optimal in some interesting identification problems. Nevertheless, even in such cases, central or interpolatory estimates may have much lower a posteriori identification errors. Moreover, it has been shown that central and interpolatory algorithms are convergent under less restrictive conditions than the least squares algorithm.

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